

# Chengsong Huang

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## EDUCATION

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### Washington University in St. Louis

Advisor: Prof. Jiaxin Huang

Ph.D. in Computer Science (GPA 4.0/4.0)

Sep. 2023 - Present

St. Louis, MO

### Fudan University

Software School of Fudan University

B.S. in Software Engineering (GPA 3.82/4.0) rank 1%

Sep. 2019 - Jun. 2023

Shanghai, China

## RESEARCH INTERESTS

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My research focuses on advancing the capabilities and accessibility of Large Language Models (LLMs). My interests include:

- **Data-Driven Self-Evolution:** Designing frameworks that enable LLMs to evolve themselves through the data and feedback they generate. This includes self-evolving training loops that synthesize their own training data and curricula from zero human-labeled data, self-rewarding models that turn their own reasoning into supervision, and environments that co-evolve with the agents trained in them. Representative works: [R-Zero](#) (ICLR 2026), a reasoning LLM that self-evolves from zero data, and [EnvHarness](#) (Google Research), which awakens static environments for agent learning.
- **Efficient LLMs:** Developing and analyzing methods to make LLMs more computationally efficient. My experience includes work on parameter-efficient fine-tuning (PEFT), speculative decoding, and adaptive test-time scaling to accelerate training and inference.

## PREPRINTS (FIRST / CO-FIRST AUTHOR)

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*\* stands for equal contribution or alphabetical ordering.*

### Data-Driven Self-Evolution

- **Huang, C.**, Wang, Z., Han, R., Yan, J., Chen, Y., CuiZhu, Z., Jiang, K., Xia, P., Yu, H., Zhuang, Y., Ming, Y., Pan, J., Dalvi Mishra, B., Huang, J., Gokturk, B., Pfister, T. and Lee, C.-Y. “[EnvHarness: Awakening Static Worlds for Agent Learning](#)”. arXiv preprint. 2026.

### Others

- Qian, Q.\*, **Huang, C.\***, Xu, J., Lv, C., Wu, M., Liu, W., Wang, X., Wang, Z., Huang, Z., Tian, M., Xu, J., Hu, K., Wang, H.-D., Hu, Y., Huang, X. and Zheng, X. “[Benchmark<sup>2</sup>: Systematic Evaluation of LLM Benchmarks](#)”. arXiv preprint. 2026.
- Gu, C.\*, **Huang, C.\***, Zheng, X., Chang, K.-W. and Hsieh, C.-J. “[Watermarking Pre-trained Language Models with Backdooring](#)”. arXiv preprint. 2022.

## PUBLISHED PAPERS (FIRST / CO-FIRST AUTHOR)

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For a complete list of publications, please visit my [Google Scholar profile](#).

*\* stands for equal contribution or alphabetical ordering.*

### Data-Driven Self-Evolution

- **Huang, C.**, Yu, W., Wang, X., Zhang, H., Li, Z., Li, R., Huang, J., Mi, H. and Yu, D. “[R-Zero: Self-Evolving Reasoning LLM from Zero Data](#)”. ICLR 2026.
- **Huang, C.**, Liu, H., Zheng, T., Dai, R., Huang, L., Li, J., Li, Z., Wei, Z., Meng, Y. and Huang, J. “[G-Zero: Self-Play for Open-Ended Generation from Zero Data](#)”. NeurIPS 2026 (**Spotlight**).
- He, Y.\*, **Huang, C.\***, Li, Z.\*, Huang, J. and Yang, Y. “[VisPlay: Self-Evolving Vision-Language Models from Images](#)”. CVPR 2026.

## ***Efficient LLMs***

- Zheng, T.\* , **Huang, C.\*** , Dai, R.\* , He, Y., Liu, R., Ni, X., Bao, H., Wang, K., Zhu, H., Huang, J., Huang, F. and Huang, H. “[Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing](#)”. ICML 2026.
- **Huang, C.**, Huang, L., Leng, J., Liu, J. and Huang, J. “[Efficient Test-Time Scaling via Self-Calibration](#)”. ICLR 2026.
- **Huang, C.**, Zheng, T., Huang, L., Li, J., Liu, H. and Huang, J. “[RelayLLM: Efficient Reasoning via Collaborative Decoding](#)”. Findings of EMNLP 2026.
- **Huang, C.**, Huang, L. and Huang, J. “[Divide, Reweight, and Conquer: A Logit Arithmetic Approach for In-Context Learning](#)”. EACL 2026.
- **Huang, C.\*** , Liu, Q.\* , Lin, B. Y.\* , Pang, T., Du, C. and Lin, M. “[LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition](#)”. COLM 2024.

## ***Others***

- Lin, B. Y.\* , **Huang, C.\*** , Liu, Q., Gu, W., Sommerer, S. and Ren, X. “[On Grounded Planning for Embodied Tasks with Language Models](#)”. AAAI 2023.

## **EXPERIENCE**

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**Google Cloud AI** Feb. 2026 - present

Student Researcher (Mentor: Zifeng Wang)

*EnvHarness: Awakening Static Worlds for Agent Learning*

**Tencent AI Lab Seattle** Jun. 2025 - Aug. 2025

Research intern (Mentor: Wenhao Yu)

*R-Zero: Self-Evolving Reasoning LLM from Zero Data*

**Sea AI lab (SAIL)** Jan. 2023 - Jun. 2023

Research intern (Mentor: Qian Liu)

*LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition*

**USC INK lab** May. 2022 - Jan. 2023

Research intern (Mentor: Xiang Ren, Bill Yuchen Lin)

*On Grounded Planning for Embodied Tasks with Language Models*

## **OPEN-SOURCE IMPACT**

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- Author and maintainer of open-source research code with **2,200+ GitHub stars** in total, including [R-Zero](#) (850+ stars), [LoraHub](#) (650+ stars), [EnvHarness](#) (600+ stars), [VisPlay](#), [RelayLLM](#), [G-Zero](#) and [Self-Calibration](#).

## **SERVICE**

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**Conference Reviewer:** ARR 2024-present, ICLR 2025-present, COLM 2024-present, AAAI 2026, CVPR 2026, ICML 2026.